THE UNIVERSITY OF LAHORE
Faculty of Information Technology | Department of Software Engineering

Mobile Application Development (CS02310) - Lab Project Task 1 Report

Create the UI and Layouts - SpendWise: Personal Expense Tracker

Yellow-highlighted text in [brackets] must be replaced with your real details before submission (names, SAP IDs, repository links, screenshots, test results). Delete this note afterwards.

1 Project and submission details

App title	SpendWise - personal expense tracker
Section / Task	Section W / Task 1 (Create the UI and layouts)
Submission date	[DD-MM-2026]
Selected stack	Kotlin, Android Studio (Android app in *app/src/main*; RecyclerView and Material Components)
Group ID / Members	[Group ID] Muhammad Faisal Rehman - SAP ID 70150146 Ahmed Shahid - SAP ID 70149224
GitHub repository	https://github.com/Faisalrehman14/madproject
Branch / Commit SHA / Tag	[main] / [full commit SHA] / **task-1**

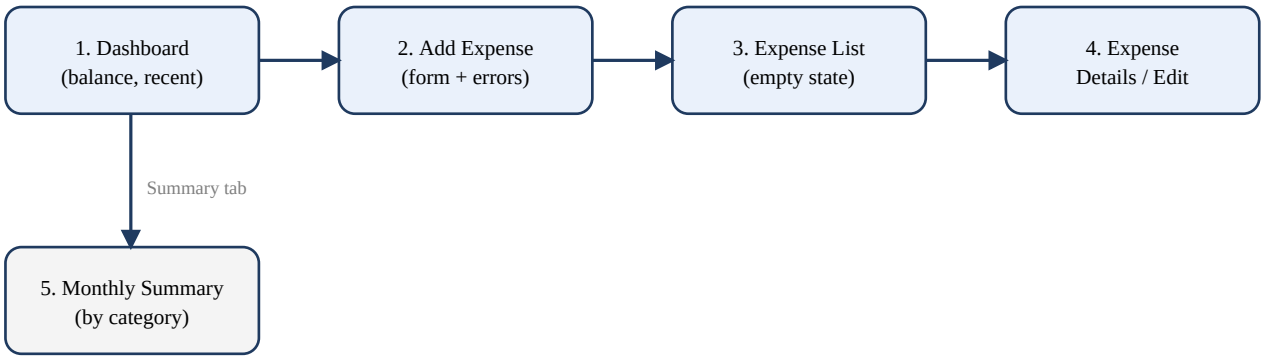
Project proposal summary

Problem: students and young earners often do not know where their money goes because they do not record small daily expenses. **Solution:** SpendWise lets a user record an expense in a few taps, see all expenses in a list and check a monthly summary by category.

Main users	Core features (3-5)
University students (primary): track pocket money and daily spending. **Young professionals**: track monthly budget.	1. Dashboard with balance and recent expenses 2. Add expense form (amount, category, date, note) 3. Expense list with category labels and empty-list design 4. Expense details / edit screen 5. Monthly summary by category

Screen-flow diagram

Main user journey: open app -> add an expense -> see it in the list -> open details -> check summary



2 What you implemented

The following screens were implemented and run on the emulator/phone using static sample data. Each screen is an Activity with its own layout file.

Screen	Files (class / layout)	What the user can do	Status
Dashboard	MainActivity.kt / activity_main.xml	See total balance and the latest expenses; buttons lead to Add Expense, List and Summary.	UI working with static data
Add Expense	AddExpenseActivity.kt / activity_add_expense.xml	Enter amount, choose category, pick date, add note; inline error text for empty or invalid amount.	UI + validation working; saving is a placeholder
Expense List	ExpenseListActivity.kt, ExpenseAdapter.kt / activity_expense_list.xml, item_expense.xml	Scroll a RecyclerView of expenses (title, category, amount, date); empty-list message when no data.	Working with static list
Expense Details	ExpenseDetailActivity.kt / activity_expense_detail.xml	View one expense; Edit and Delete buttons.	UI working; actions are placeholders
Monthly Summary	SummaryActivity.kt / activity_summary.xml	See total spent and per-category amounts for the month.	Static sample values

Layout adaptation: every screen uses ConstraintLayout / ScrollView so no control is clipped. Landscape variants are provided in *res/layout-land* for Dashboard and Add Expense (two-column layout). Tested on two sizes: [e.g. Pixel 4 - 5.7 in] and [e.g. Pixel Tablet / small 4.0 in phone], in portrait and landscape.
Readability: minimum 16sp body text, 48dp touch targets, dark text on light surface (contrast above 4.5:1), consistent 16dp margins, one primary colour for actions and red/green only with a text label (not colour alone).
Known limitations: no database, expenses are not saved after restart, no charts or login. These are planned for later tasks.

Screenshots (labelled)

Paste screenshot
Fig 1: Dashboard

Paste screenshot
Fig 2: Add Expense

Paste screenshot
Fig 3: List

Paste screenshot
Fig 4: Details

Paste screenshot
Fig 5: Empty list

Paste screenshot
Fig 6: Summary

Paste screenshot
Fig 7: Landscape

Paste screenshot
Fig 8: 2nd device

Short screen recording (2-3 min): [video link or file name]

3 Activities you performed

Step	Activity and reasoning	Tool	Done by
1	Planning: chose users, features and screen list; drew the screen-flow. Kept one main journey (add and review an expense) so the app is quick to use.	draw.io / paper	[names]
2	Created the Android Studio project (Kotlin) and pushed it to the shared GitHub repository.	Android Studio, Git	[names]
3	Built layouts in XML. Used ConstraintLayout for flexible resizing and RecyclerView so long expense lists scroll smoothly.	Layout Editor	[names]
4	Connected controls: buttons open the next Activity via Intent; form validation shows an error under the amount field; sample expenses kept in a Kotlin list.	Kotlin	[names]
5	Added the empty-list design so the user is told to add the first expense instead of seeing a blank screen.	Kotlin / XML	[names]
6	Tested on emulator and phone in portrait and landscape; fixed spacing and clipping; tagged **task-1**.	AVD / USB phone	[names]

4 Issues faced and how you tackled them

Record only problems you genuinely met. Example row below shows the expected style; replace it with your own real issues, or write that no significant issue occurred and list the checks you performed.

Issue or symptom	Investigation and cause	Action and verification
[Describe what went wrong, on which screen]	[How you reproduced it and the cause]	[Fix, retest result, resolved or not]
[Issue 2 (if any)]	[...]	[...]

5 Testing and individual contributions

Test	Expected result	Actual result
Save with empty amount	Error message under the field; form stays open	[Pass/Fail]
Open Add Expense and choose category	Selected category is shown on the form	[Pass/Fail]
Open list with no data	Empty-state message and Add Expense button	[Pass/Fail]
Tap an expense in the list	Details screen opens for that expense	[Pass/Fail]
Open Monthly Summary	Total and category amounts visible	[Pass/Fail]
Rotate to landscape	All controls visible, content scrolls, no clipping	[Pass/Fail]
Second screen size	Layout remains usable	[Pass/Fail]

Member	Activity	File / feature	Commit link(s)
Muhammad Faisal Rehman 70150146	[e.g. Dashboard, Add Expense UI]	[MainActivity, AddExpenseActivity]	[commit URL]
Ahmed Shahid 70149224	[e.g. List, Details, Summary UI]	[ExpenseListActivity, SummaryActivity]	[commit URL]

Pair-programming: [state who paired with whom and on what]

6 GitHub repository and version control

Repository URL	https://github.com/Faisalrehman14/madproject
Submitted commit link	https://github.com/Faisalrehman14/madproject/commit/[SHA]
Tag / release	task-1 [link to tag after you create it]
Branching and merging	[e.g. each member worked on a feature branch, merged to main by pull request]

README	Required tools (Android Studio, JDK), run steps and known limitations [add README.md to repo]. Repository is public, so lecturer access is not needed.

SLATE package: MAD_W_[GroupID]_Task1_Report.pdf, repository and commit links, screenshots, screen-flow diagram and 2-3 minute recording. Verify the report, recording and tagged commit show the same version.